

CONSTRUCTION OF A PANGENOME TO ADVANCE ALMOND (*P. DULCIS*) GENETICS AND BREEDING

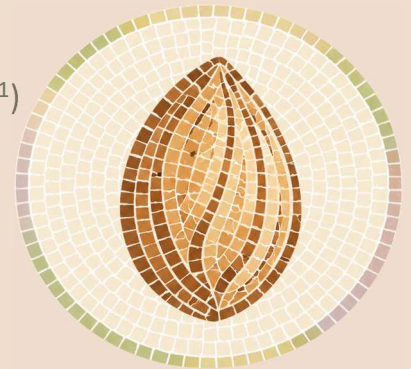
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WHY AN ALMOND PANGENOME?

Almond genomics today

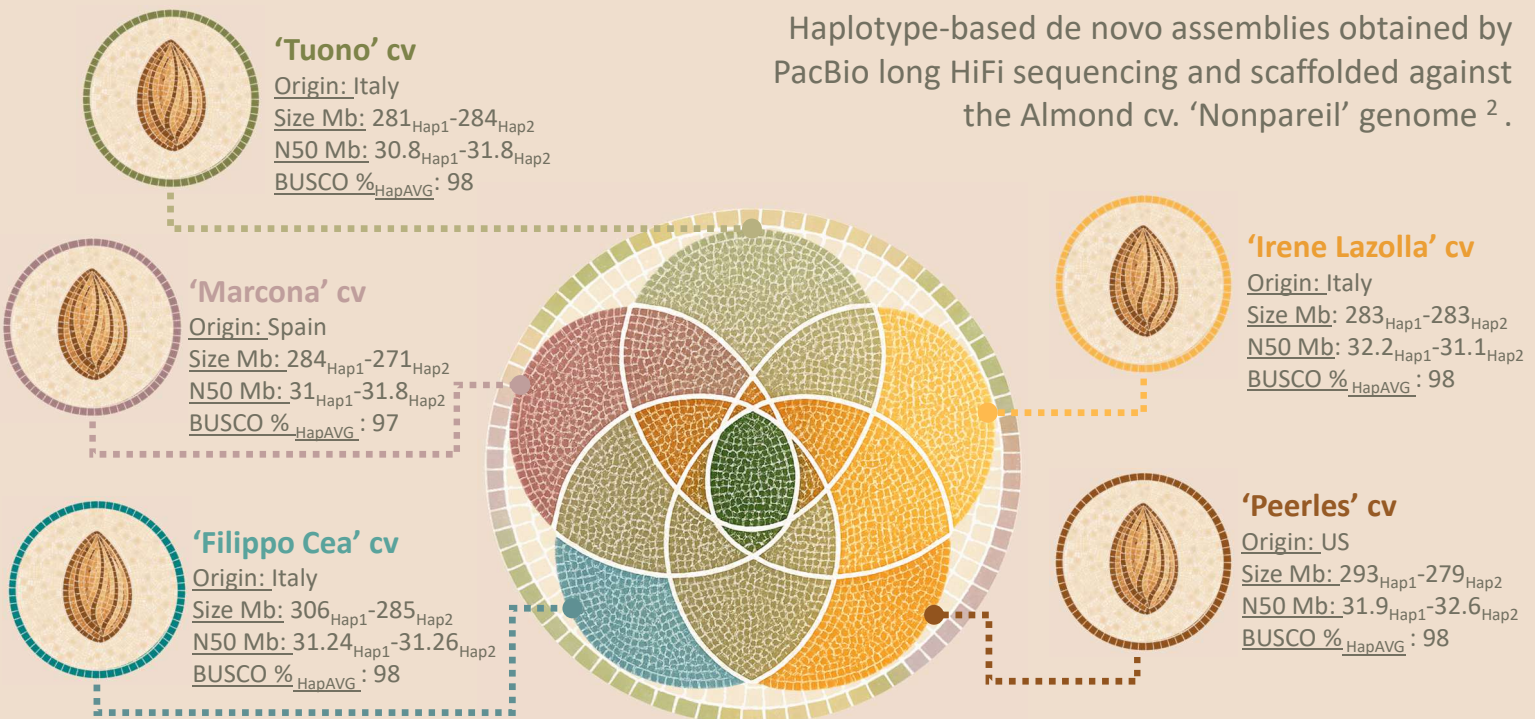
- Almond (*Prunus dulcis*, $2x=2n=16$) is a globally relevant nut crop (~ 3.9 Mt ¹)
- Current resources rely on few reference genomes
- Almond displays extensive structural and allelic diversity
- Single-reference approaches miss relevant genomic variation
- A pangenome framework is required to capture species-wide diversity



THE GENOMIC RESOURCES

Five almond assemblies generated within our research group

Haplotype-based de novo assemblies obtained by PacBio long HiFi sequencing and scaffolded against the Almond cv. 'Nonpareil' genome ².



The future almond pangenome framework

1: FAOSTAT, 2024 www.fao.org ;2: NCBI BioProject ID [PRJNA769745](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA769745)

ABOUT ME

I am currently a Postdoctoral Researcher at the University of Bari Aldo Moro (Italy), working in the group of Prof. Stefano Pavan, who contributed to the publication of the first almond reference genome. My research interests focus on plant population genomics and genotype-phenotype association studies.

